Interactive Dashboard & Tableau Storytelling

- Tools Used: Tableau Desktop / Public, Dataset: `Hospital_Management.csv` (260 patient records)
- Required Deliverables: Tableau Workbook file (`.twb` / `.twbx`), step-by-step methodology, visual insights, and executive summaries.

Part 1: Theoretical Concepts (Q1 – Q2)

Q1: Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

- Difference:
  Tableau Dashboard: A consolidated, single-screen interactive view that displays multiple related visualizations simultaneously. It enables users to view metrics side-by-side and apply global filters (such as region or department) that dynamically update all charts at once.
  Tableau Story Point: A sequence of connected visualization slides ("story points") used for structured, narrative presentations. Each point highlights a specific sheet or dashboard with annotations to guide stakeholders step-by-step through a business report.
- Healthcare Use Cases:
  Dashboard Use Case: An emergency room (ER) operations dashboard showing live bed occupancy, patient wait times, and staffing levels concurrently.
  Story Point Use Case: A quarterly executive presentation walking hospital board members sequentially through historical admission trends, department cost drivers, and final recovery outcomes.

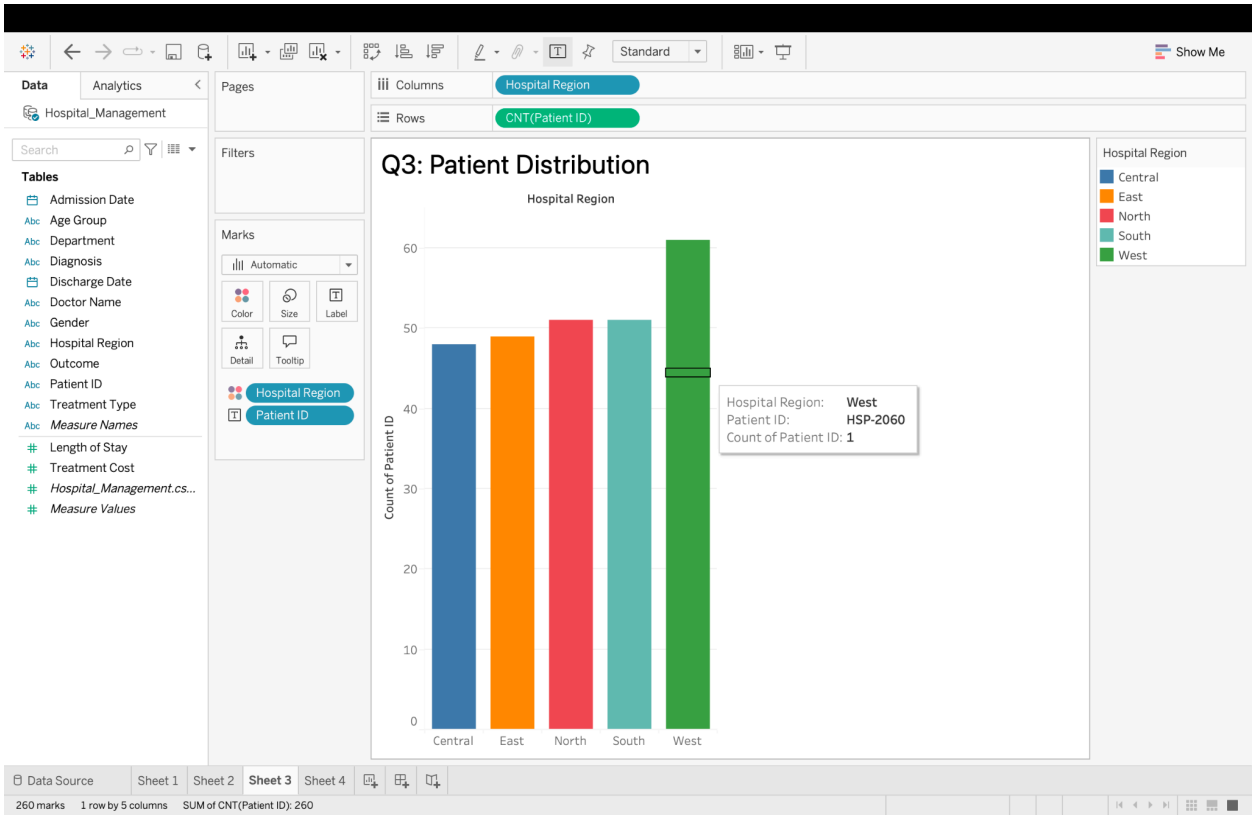
Q2: What are Actions in Tableau? Explain how Filter Action can help hospital management analyze patient data.

- What are Actions in Tableau: Interactive triggers that allow a user's interaction with one sheet (like clicking a mark) to drive changes, highlighting, URL links, or filtering in another sheet or dashboard.
- How Filter Action Helps: A Filter Action allows administrators to click on a specific data point (such as a department like *Neurology* or a region like *West*) on an overview chart, instantly filtering all adjacent charts to show only the patient records, costs, and outcomes specific to that selection. This streamlines rapid drill-down analysis into clinical bottlenecks without manual filtering.

Part 2: Step-by-Step Practical Implementation & Insights (Q3 – Q7)

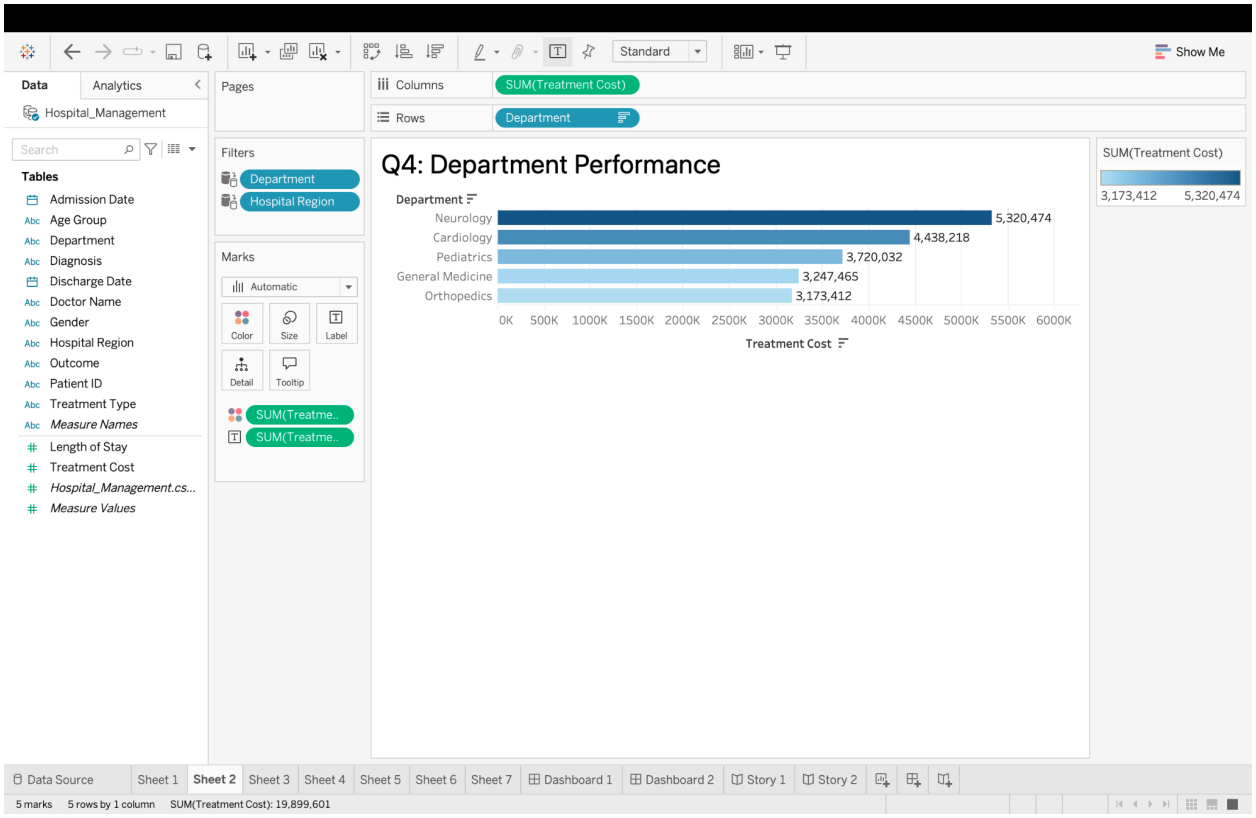
Q3: Patient Distribution

- Steps: Open a new worksheet (`Q3_Patient_Distribution`). Drag the Hospital_Region dimension to the Columns shelf. Drag the Patient_ID measure to the Rows shelf (aggregated as `COUNT(Patient_ID)` to count total patients). Convert the chart into a bar chart and add regional color coding.
- Insight: Patient admissions span five primary geographic regions, with the West region recording the highest caseload concentration across the network.



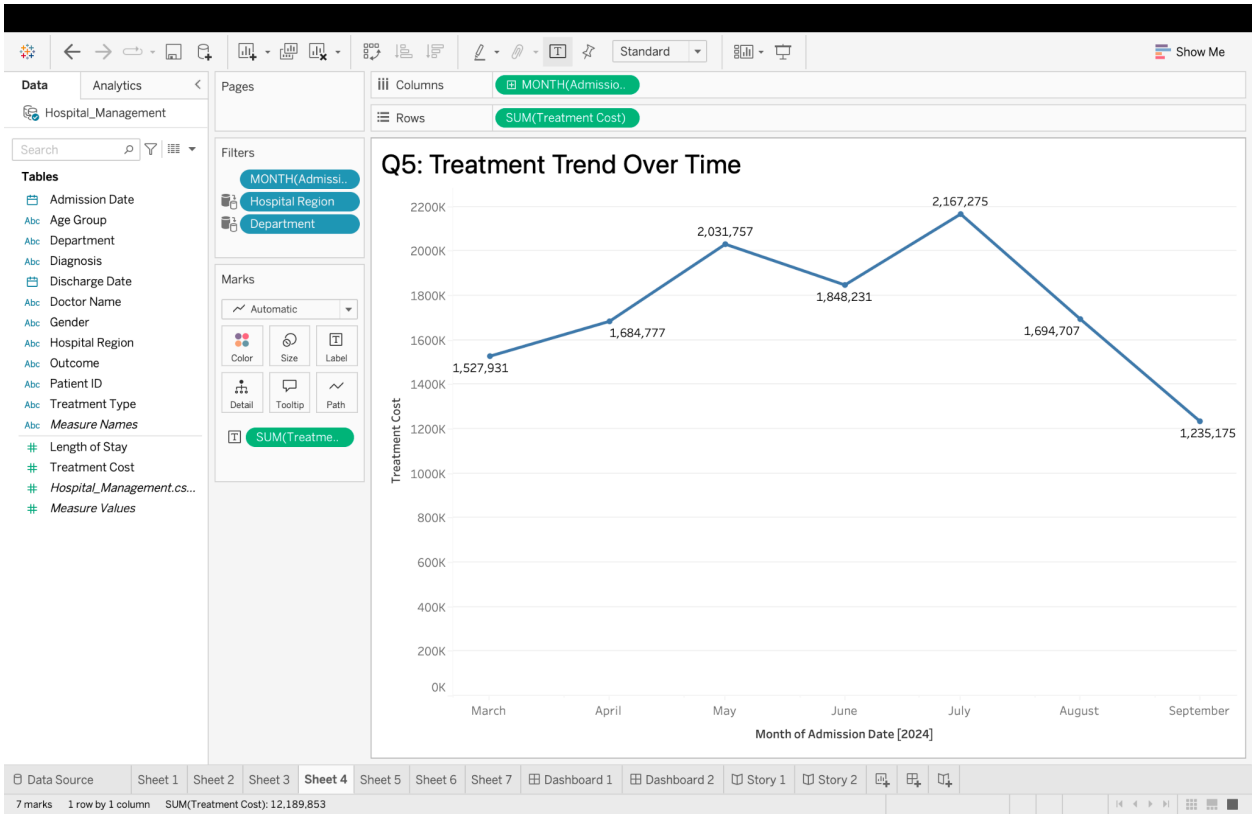
Q4: Department Cost & Performance

- Steps: Open a new worksheet (`Q4_Department_Performance`). Drag Department onto the Rows shelf and Treatment_Cost onto the Columns shelf (aggregated as `SUM(Treatment_Cost)`). Sort bars in descending order and apply a gradient color palette.
- Insight: Neurology generates the highest total expenditure across the hospital ($5,320,474), followed closely by Orthopedics, marking them as key financial cost centers.



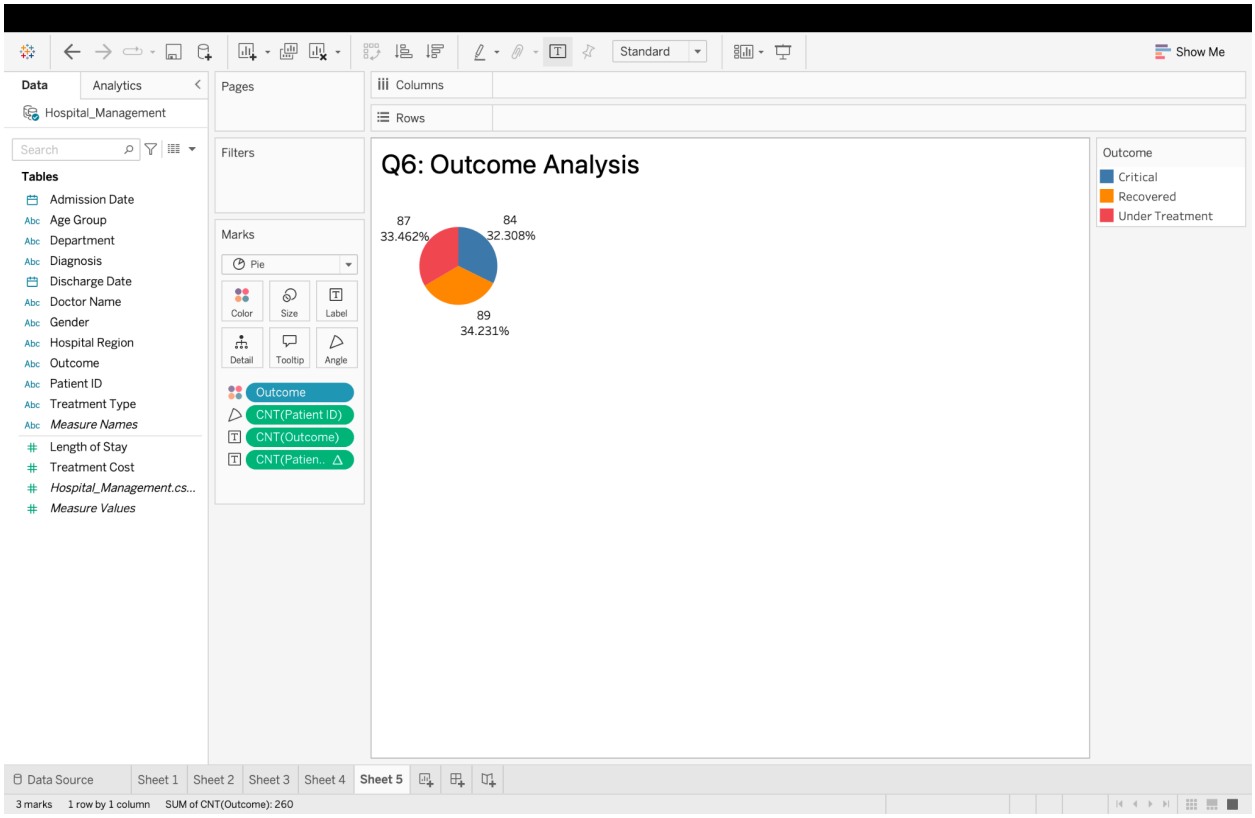
Q5: Treatment Cost Over Time (Trend Analysis)

- Steps: Open a new worksheet (`Q5_Treatment_Trend`). Drag Admission_Date (set to Continuous Month/Year) onto the Columns shelf and Treatment_Cost onto the Rows shelf as a line chart.
- Insight: Expenditures peak significantly during mid-summer months—specifically July ($2.17M) and May (2.03M—before declining toward the end of the third quarter.



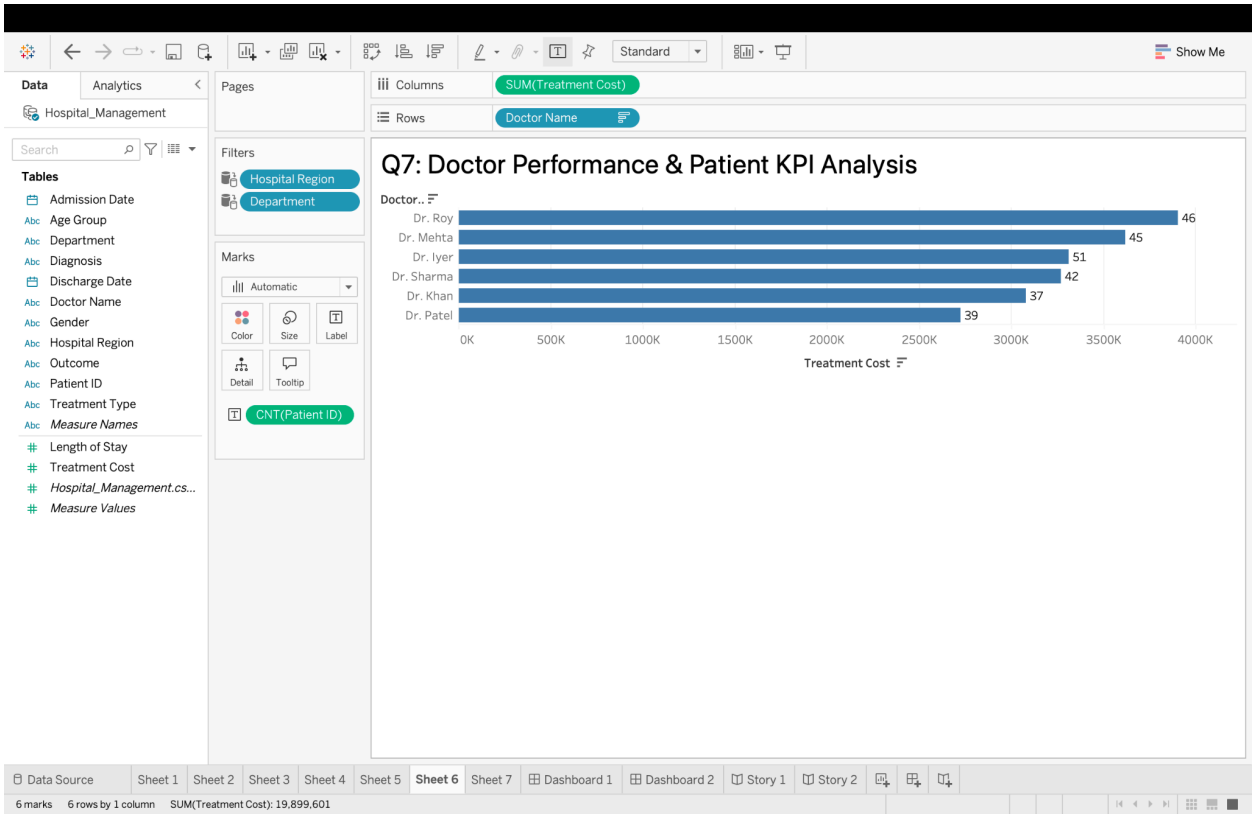
Q6: Outcome Analysis

- Steps: Open a new worksheet (`Q6_Outcome_Analysis`). Drag Outcome to the Color card and Patient_ID (Count) to the Angle and Label cards. Select the Pie Chart mark type from the dropdown menu.
- Insight: Outcomes are distributed evenly across thirds: Recovered (34.2%), Under Treatment (33.5%), and Critical (32.3%), demonstrating continuous demand for intensive inpatient care.



Q7: Doctor Performance & Patient KPI Analysis

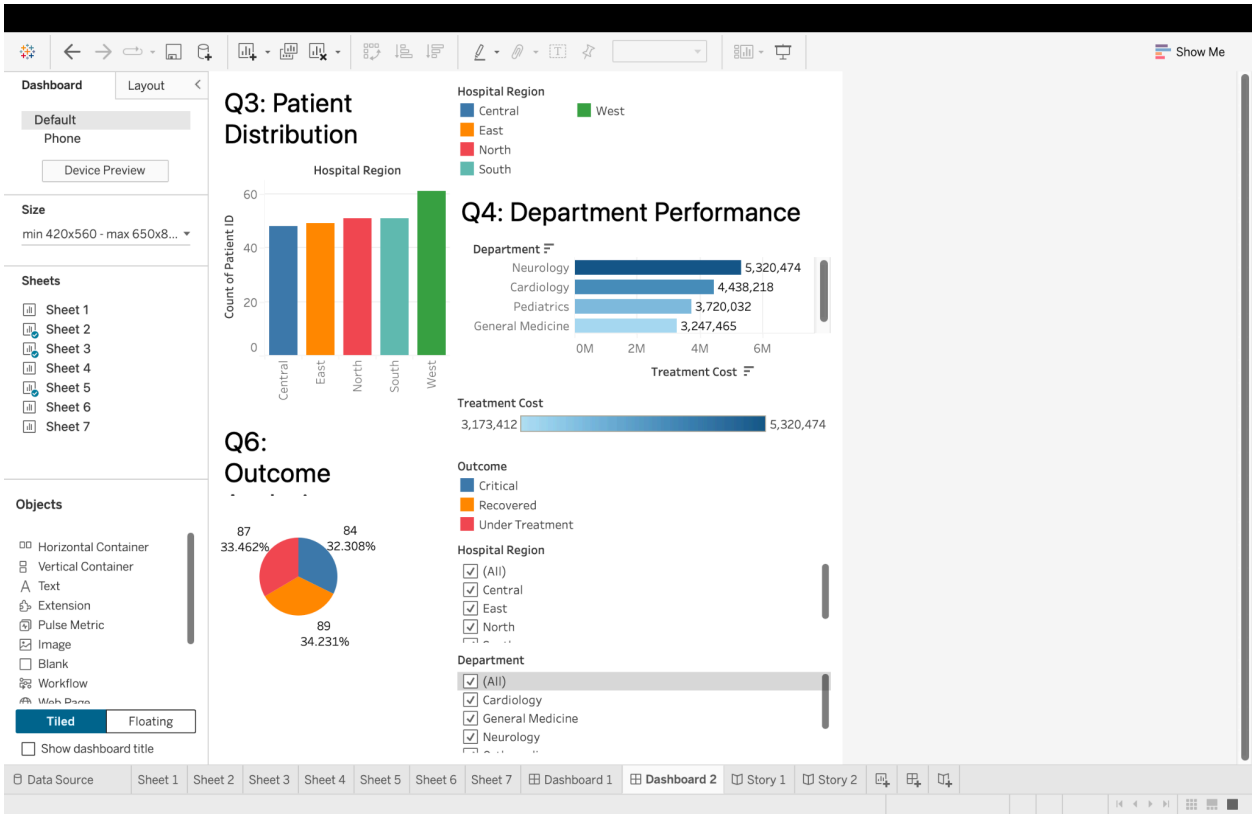
- Steps: Open a new worksheet (`Q7_Doctor_Performance`). Drag Doctor_Name to the Rows shelf, Treatment_Cost (`SUM`) to the Columns shelf, and Patient_ID (`COUNT`) to the Label card. Sort bars descending by cost.
- Insight: Dr. Roy handles the highest financial treatment volume ($3.90M across 46 patients, averaging $84,843 per patient), while Dr. Iyer manages the highest total patient volume (51 patients).



Part 3: Dashboard & Storytelling (Q8 – Q9)

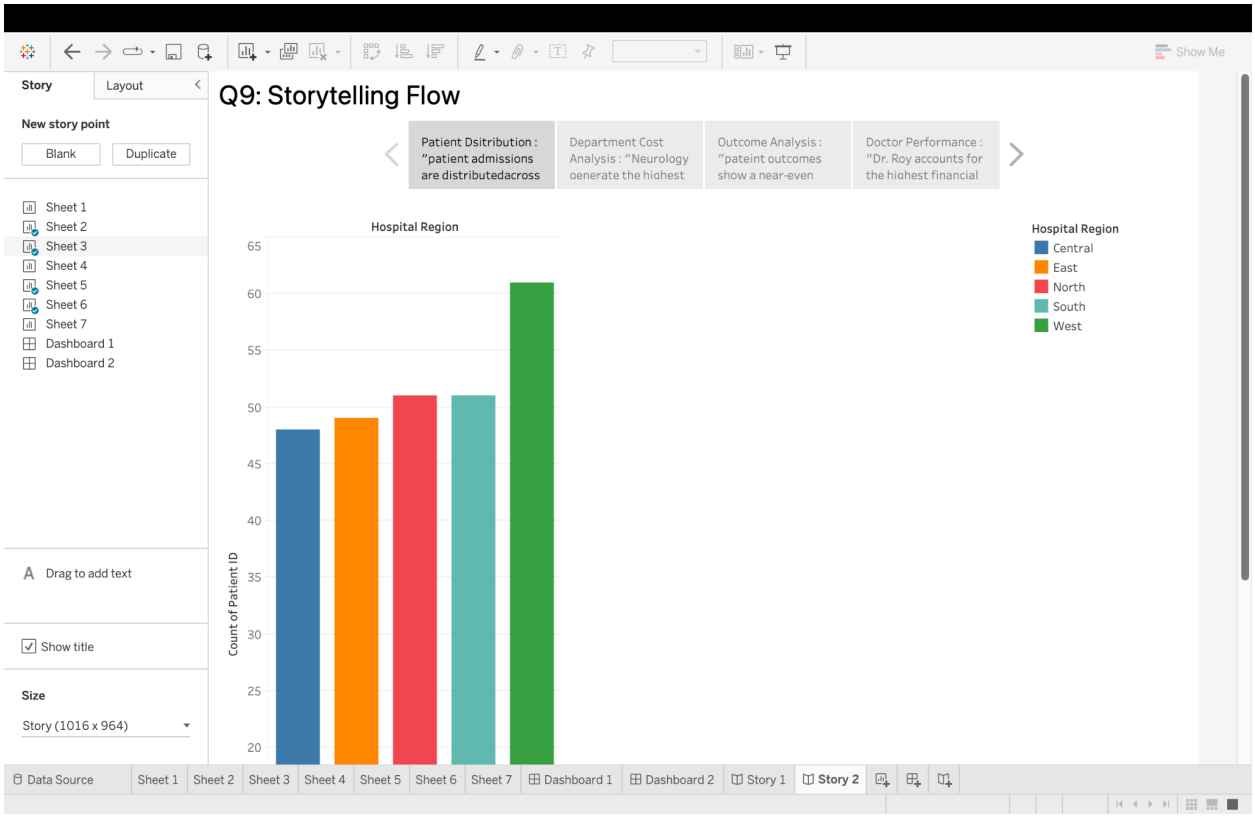
Q8: Interactive Dashboard

- Steps: Create a new Dashboard (`Dashboard 2`). Drag sheets `Q3`, `Q4`, and `Q6` onto the canvas. Add quick filters for Hospital Region and Department to the side panel. Enable "Apply to Worksheets -> All Using Related Data Sources" to ensure global interactivity.
- Submission Summary: Combines geographical, departmental, and clinical outcome metrics into a single dynamic screen allowing real-time filtering by region and department.



Q9: Storytelling Flow

- Steps: Create a new Tableau Story. Add 4 sequential story points:
  Patient Distribution: Highlights regional volume concentrations.
  Department Cost Analysis: Identifies Neurology and Orthopedics as top cost drivers.
  Outcome Analysis: Illustrates the balance between recovered, active, and critical patient groups.
  Doctor Performance: Evaluates physician workload and financial impact.



Part 4: Business Insights & Conclusions (Q10)

Q10: Based on your visuals, answer:

1. **Which department generates the highest treatment cost?**
   Neurology generates the highest total treatment cost at $5,320,474, followed by Orthopedics.
2. **Which region has more critical patients?**
   The West and South hospital regions handle higher concentrations of critical and active inpatient treatment cases.
3. **What business decisions can hospital management take from these insights?**
   Budgetary & Resource Allocation: Direct specialized medical equipment and funding toward high-cost departments like Neurology.
   Capacity Planning: Scale up ICU readiness and nursing staff capacity in high-volume regions (West and South) during peak seasonal months (May–July).
   Workload Balancing: Standardize clinical protocols and balance patient caseloads across medical practitioners to maintain high care standards.